ESOFT METRO CAMPUS

Bachelor of Information Technology (Honus) Degree

Batch 002

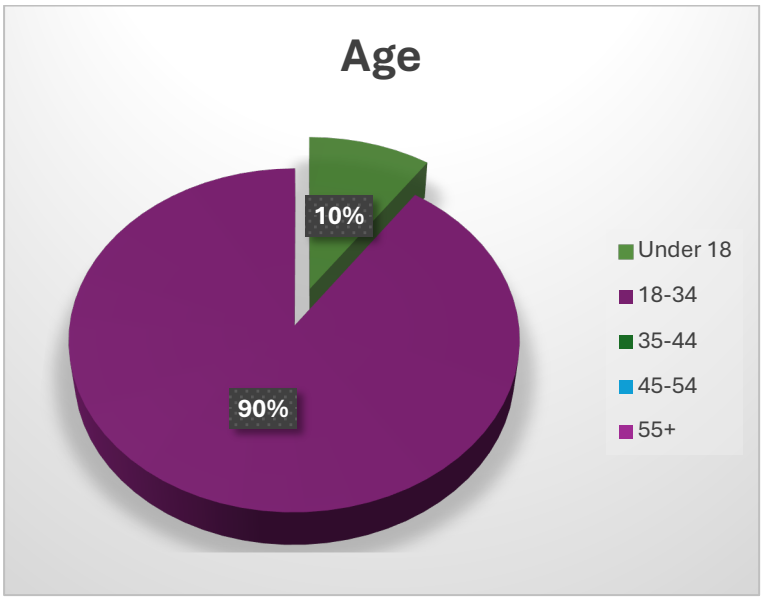
Level 4 – Semester 2

CCU4303 – Research Methodology for IT Professionals
Assignment –Questionnaire _B002

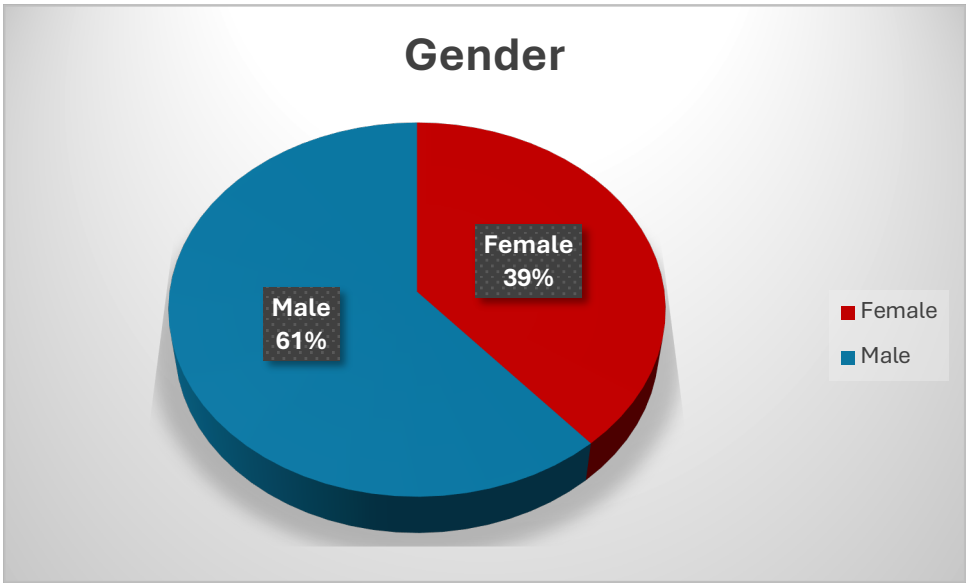
Name : R.M.S.Nethmini Rathnayaka

ESOFT-ID : E123375

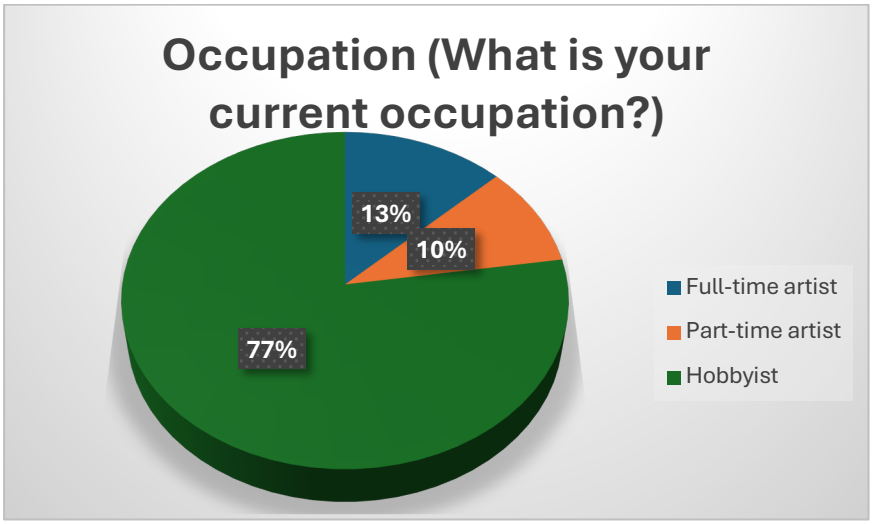
Age	Sum
Under 18	3
18-34	28
35-44	0
45-54	0
55+	0



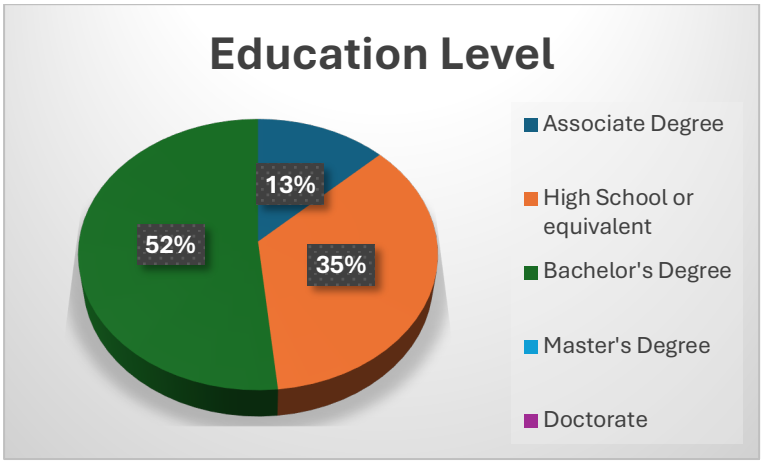
Gender	Count
Female	12
Male	19



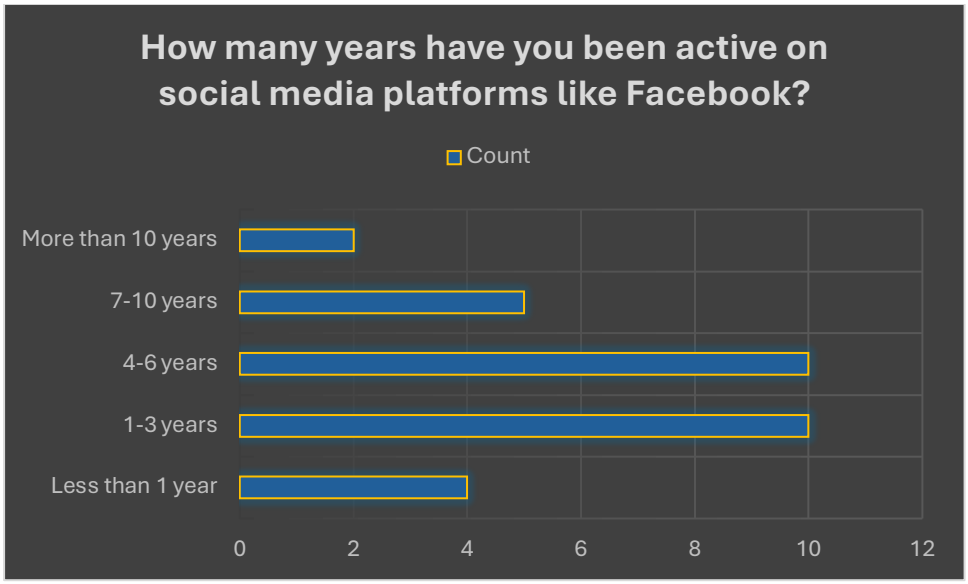
Occupation (What is your current occupation?)	Count
Full-time artist	4
Part-time artist	3
Hobbyist	24



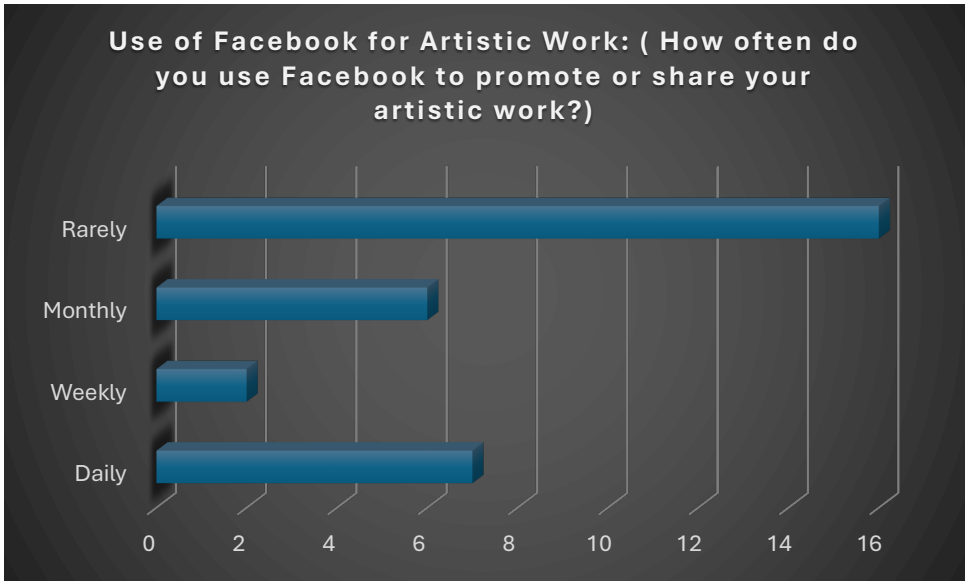
Education Level	Count
Associate Degree	4
High School or equivalent	11
Bachelor's Degree	16
Master's Degree	0
Doctorate	0



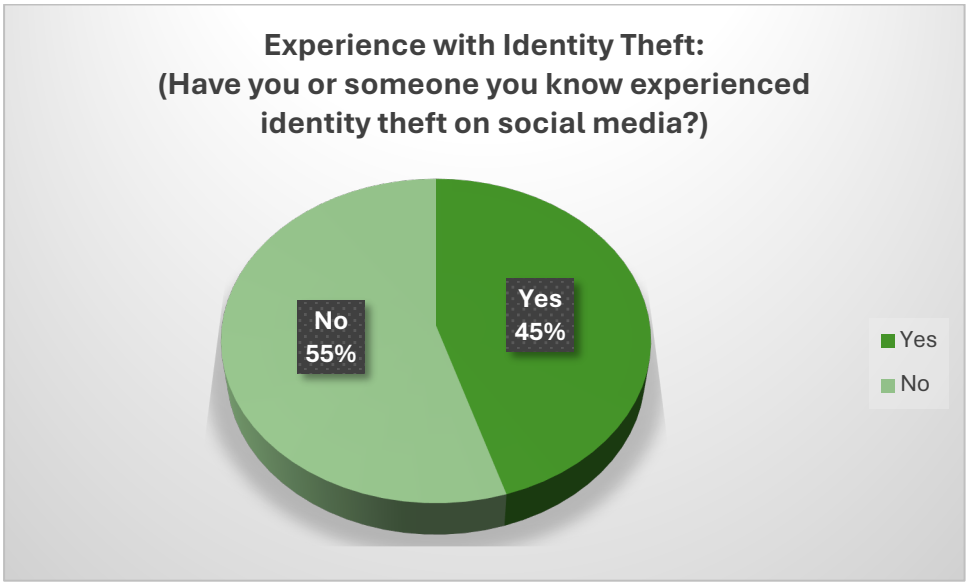
How many years have you been active on social media platforms like Facebook?	Count
Less than 1 year	4
1-3 years	10
4-6 years	10
7-10 years	5
More than 10 years	2



Use of Facebook for Artistic Work: (How often do you use Facebook to promote or share your artistic work?)	Count
Daily	7
Weekly	2
Monthly	6
Rarely	16



Experience with Identity Theft: (Have you or someone you know experienced identity theft on social media?)	Count
Yes	14
No	17

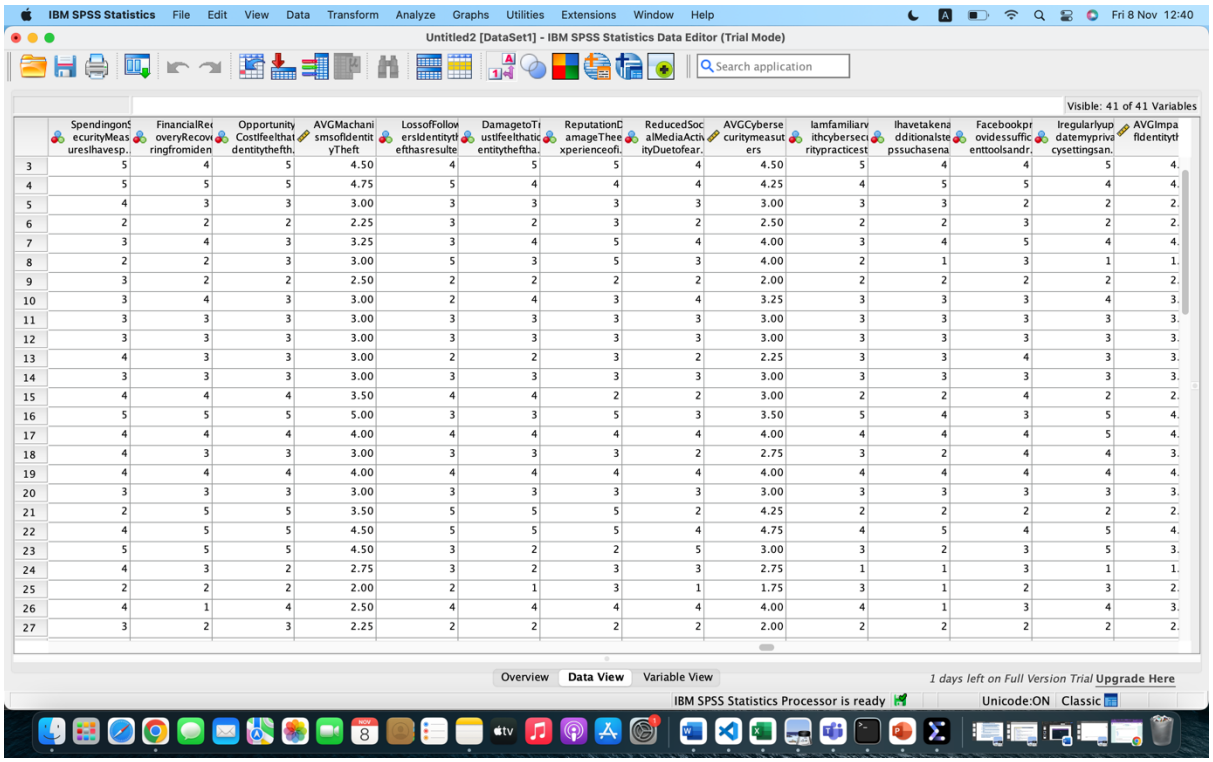


Step 03- Inferential Statistics for the Variable based question

a) Using Pilot study(40) analysis

in here we check the reliability and validity in the questionnaire

first open the data set in SPSS

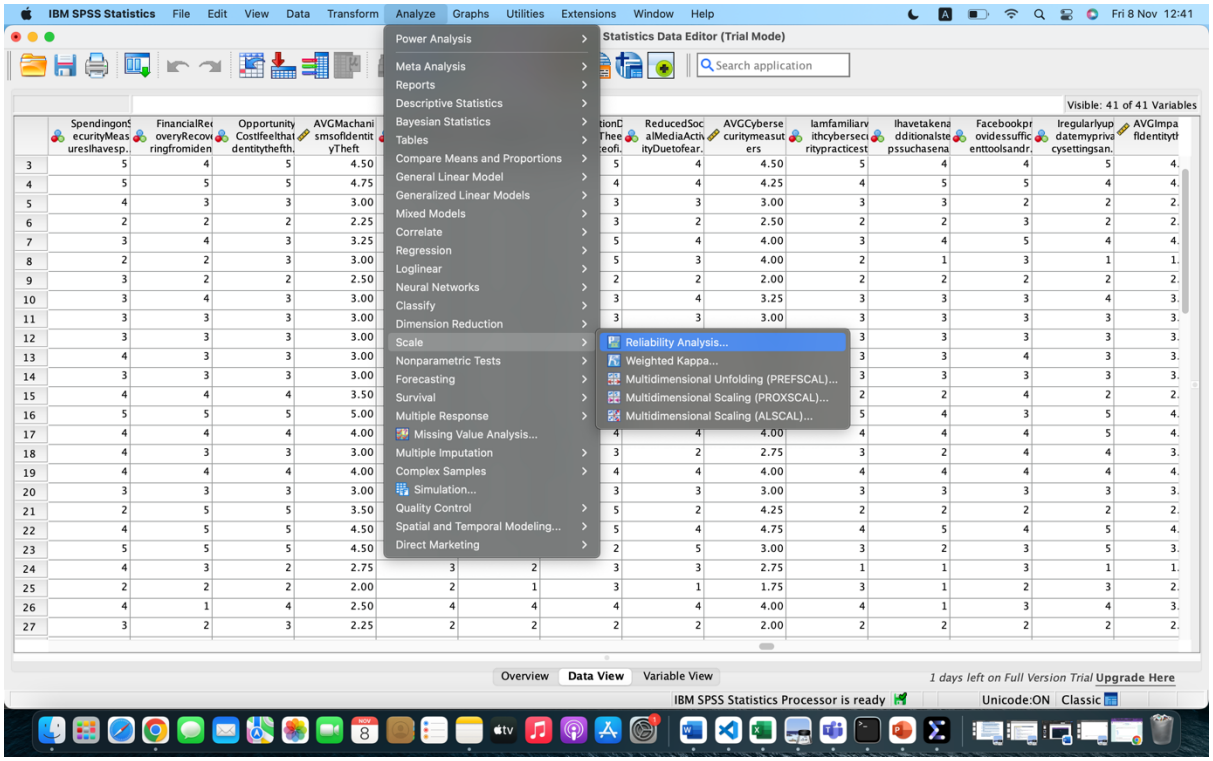


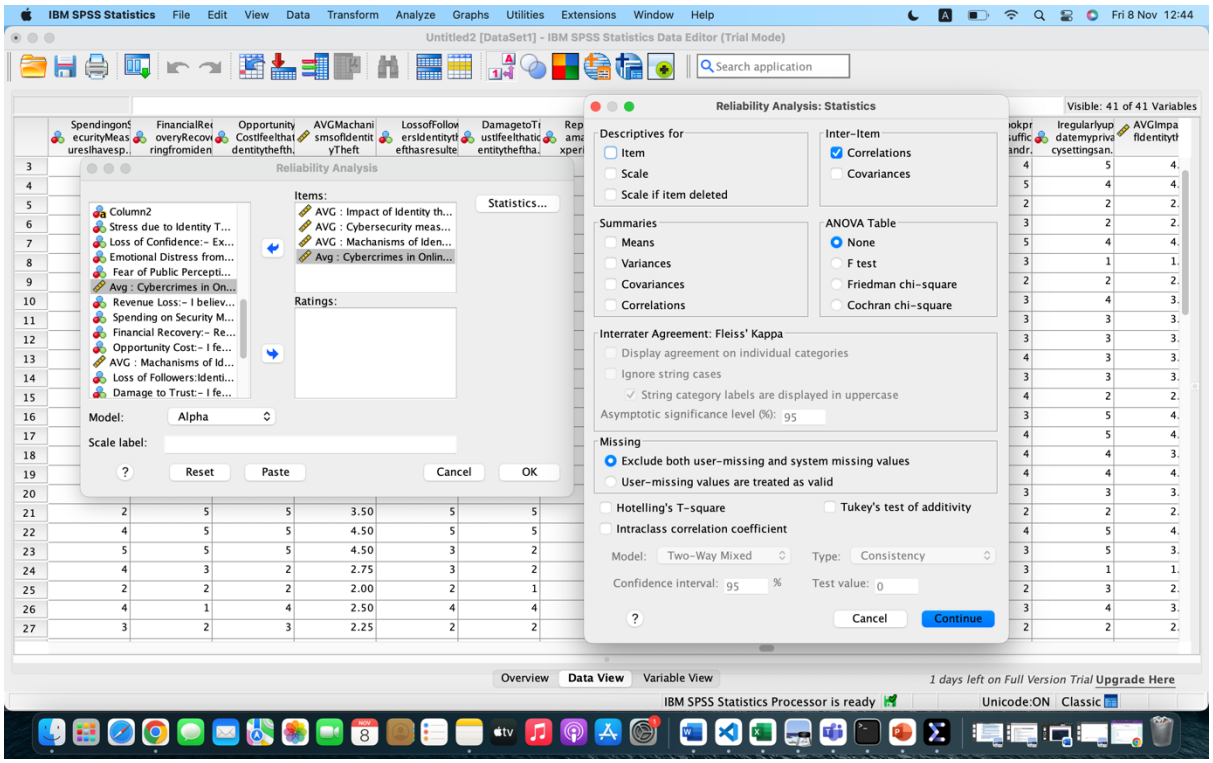
	SpendingonSecurityMeasuresIhavesp...	FinancialRecoveryRecoveringfromiden	Opportunity Costifeelthatidentitytheftth...	AVGMachanismsofIdentityTheft	LossofFollowersIdentityTefthasresulte	DamagetoTrustifeelthatidentitythefta...	ReputationDamageTheExperienceofi...	ReducedSocialMediaActivityDuetofear...	AVGCybersecuritymeasuters	IamfamiliarwithcybersecuritypracticesI	IhavetakenadditionalstepssuchasenaI	FacebookprovidessufficenttoolsandI	IregularlyupdatemyprivacysettingsaI	AVGImpactofIdentityTh
3	5	4	5	4.50	4	5	5	4	4.50	5	4	4	5	4.
4	5	5	5	4.75	5	4	4	4	4.25	4	5	5	4	4.
5	4	3	3	3.00	3	3	3	3	3.00	3	3	2	2	2.
6	2	2	2	2.25	3	2	3	2	2.50	2	2	3	2	2.
7	3	4	3	3.25	3	4	5	4	4.00	3	4	5	4	4.
8	2	2	3	3.00	5	3	5	3	4.00	2	1	3	1	1.
9	3	2	2	2.50	2	2	2	2	2.00	2	2	2	2	2.
10	3	4	3	3.00	2	4	3	4	3.25	3	3	3	4	3.
11	3	3	3	3.00	3	3	3	3	3.00	3	3	3	3	3.
12	3	3	3	3.00	3	3	3	3	3.00	3	3	3	3	3.
13	4	3	3	3.00	2	2	3	2	2.25	3	3	4	3	3.
14	3	3	3	3.00	3	3	3	3	3.00	3	3	3	3	3.
15	4	4	4	3.50	4	4	2	2	3.00	2	2	4	2	2.
16	5	5	5	5.00	3	3	5	3	3.50	5	4	3	5	4.
17	4	4	4	4.00	4	4	4	4	4.00	4	4	4	5	4.
18	4	3	3	3.00	3	3	3	2	2.75	3	2	4	4	3.
19	4	4	4	4.00	4	4	4	4	4.00	4	4	4	4	4.
20	3	3	3	3.00	3	3	3	3	3.00	3	3	3	3	3.
21	2	5	5	3.50	5	5	5	2	4.25	2	2	2	2	2.
22	4	5	5	4.50	5	5	5	4	4.75	4	5	4	5	4.
23	5	5	5	4.50	3	2	2	5	3.00	3	2	3	5	3.
24	4	3	2	2.75	3	2	3	3	2.75	1	1	3	1	1.
25	2	2	2	2.00	2	1	3	1	1.75	3	1	2	3	2.
26	4	1	4	2.50	4	4	4	4	4.00	4	1	3	4	3.
27	3	2	3	2.25	2	2	2	2	2.00	2	2	2	2	2.

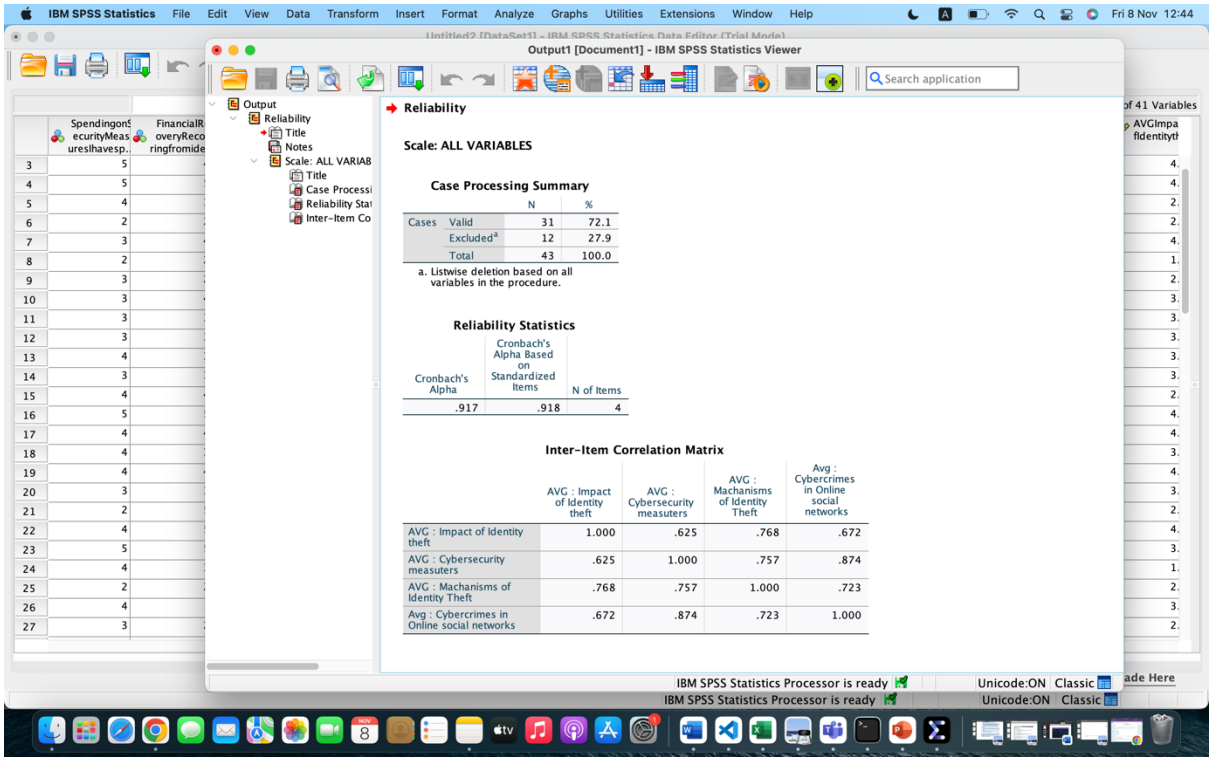
- to check the reliability test -Cronbach's alpha

Steps in reliability analysis

*Analysis >>> Scale >>> Reliability Analysis


IBM SPSS Statistics File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help
Fri 8 Nov 12:41
Statistics Data Editor (Trial Mode)
Search application
Visible: 41 of 41 Variables
Power Analysis
Meta Analysis
Reports
Descriptive Statistics
Bayesian Statistics
Tables
Compare Means and Proportions
General Linear Model
Generalized Linear Models
Mixed Models
Correlate
Regression
Loglinear
Neural Networks
Classify
Dimension Reduction
Scale
Nonparametric Tests
Forecasting
Survival
Multiple Response
Missing Value Analysis...
Multiple Imputation
Complex Samples
Simulation...
Quality Control
Spatial and Temporal Modeling...
Direct Marketing
Reliability Analysis...
Weighted Kappa...
Multidimensional Unfolding (PREFSCAL)...
Multidimensional Scaling (PROXSCAL)...
Multidimensional Scaling (ALSCAL)...
Overview
Data View
Variable View
1 days left on Full Version Trial Upgrade Here
IBM SPSS Statistics Processor is ready
Unicode:ON Classic


IBM SPSS Statistics File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help
Fri 8 Nov 12:44
Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor (Trial Mode)
Search application
Visible: 41 of 41 Variables
Reliability Analysis
Column2
Stress due to Identity T...
Loss of Confidence:- Ex...
Emotional Distress from...
Fear of Public Percepti...
Avg : Cybercrimes in On...
Revenue Loss:- I believ...
Spending on Security M...
Financial Recovery:- Re...
Opportunity Cost:- I fe...
AVG : Machanisms of Id...
Loss of Followers:Identi...
Damage to Trust:- I fe...
Items:
AVG : Impact of Identity th...
AVG : Cybersecurity meas...
AVG : Machanisms of Iden...
Avg : Cybercrimes in Onlin...
Ratings:
Statistics...
Model: Alpha
Scale label:
Reset Paste Cancel OK
Reliability Analysis: Statistics
Descriptives for
Item
Scale
Scale if item deleted
Inter-Item
Correlations
Covariances
Summaries
Means
Variances
Covariances
Correlations
ANOVA Table
None
F test
Friedman chi-square
Cochran chi-square
Interrater Agreement: Fleiss' Kappa
Display agreement on individual categories
Ignore string cases
String category labels are displayed in uppercase
Asymptotic significance level (%): 95
Missing
Exclude both user-missing and system missing values
User-missing values are treated as valid
Hotelling's T-square
Tukey's test of additivity
Intraclass correlation coefficient
Model: Two-Way Mixed
Type: Consistency
Confidence interval: 95 %
Test value: 0
Cancel Continue
Overview
Data View
Variable View
1 days left on Full Version Trial Upgrade Here
IBM SPSS Statistics Processor is ready
Unicode:ON Classic



Output in Cronbach's alpha Analysis

*in here answer is between 0 and 1 .

*If it is 0 - Questionnaire is **not reliable**

*If it is 1 - Questionnaire is **reliable**

*If it is > 0.7 - Questionnaire is **reliable**

*If it is 0.6 and 0.7 - Questionnaire is **Marginally accepted**

*If it is 0.5 and 0.6 - Questionnaire is **Poor**

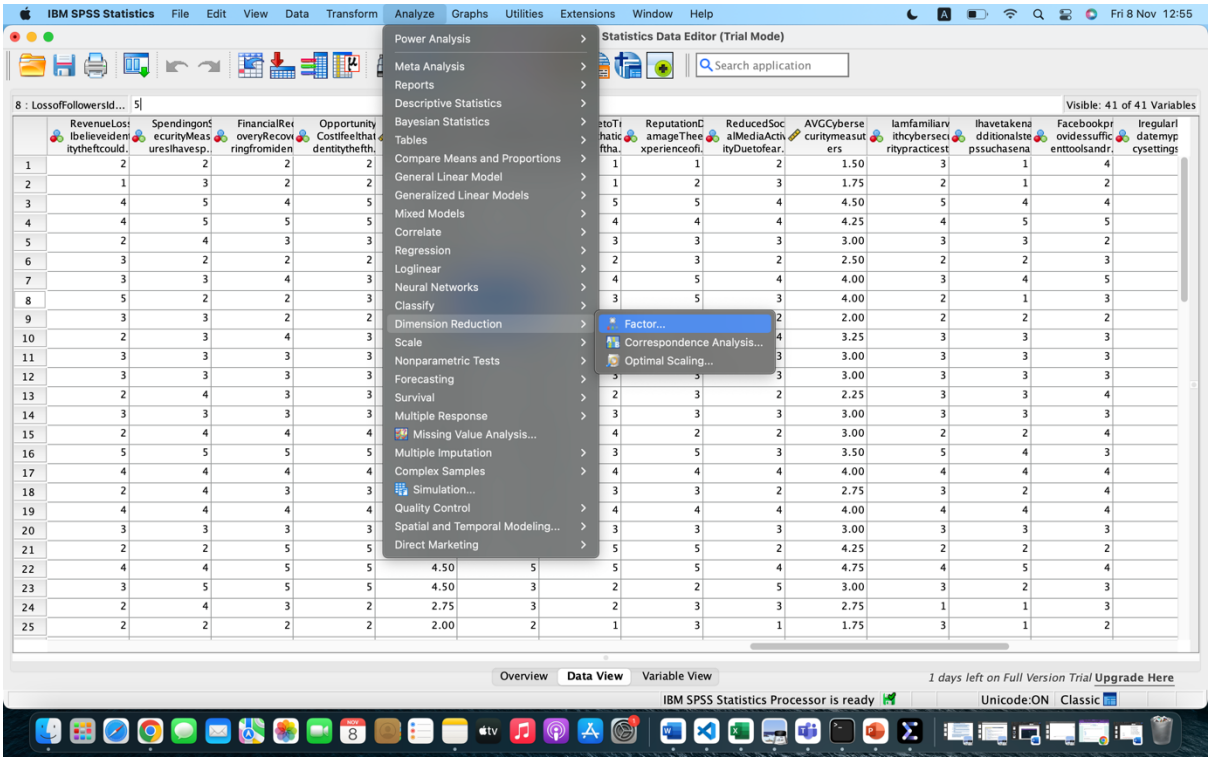
*If it is less than 0.5 - Questionnaire is **un acceptable**

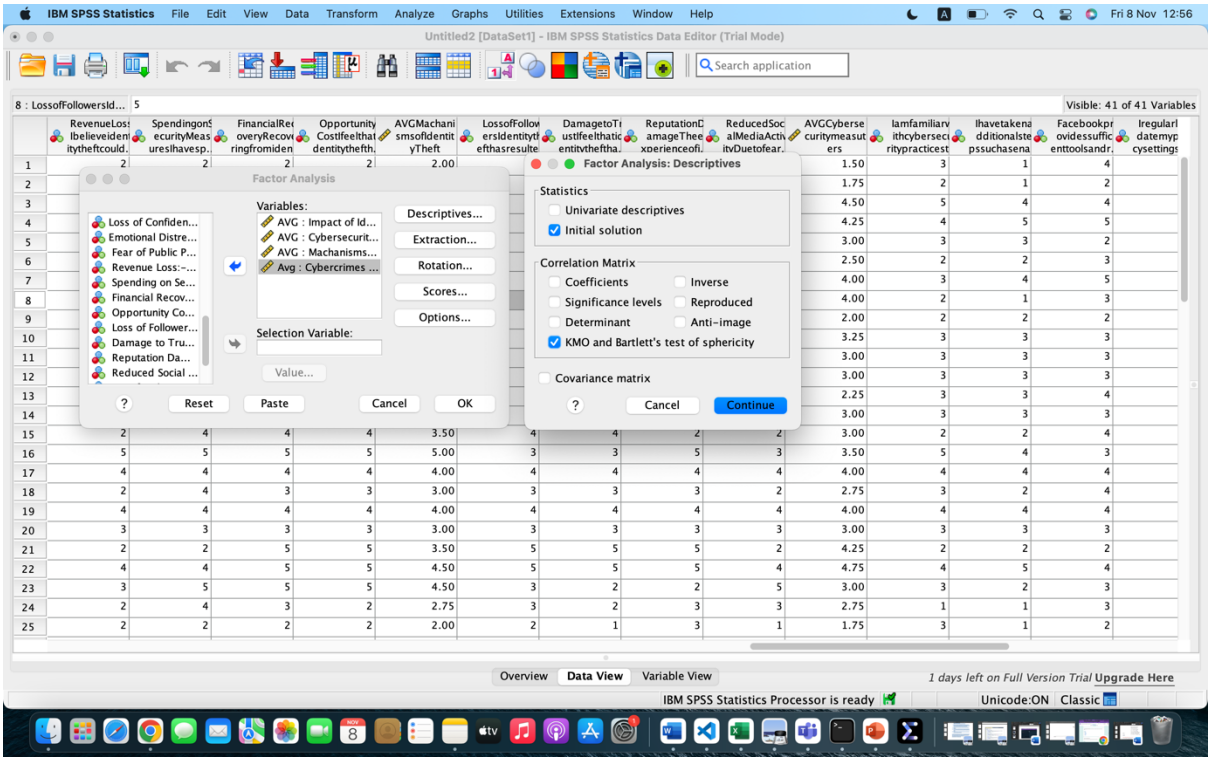
In here Cronbach's alpha value is .917 . it is >0.7 so this Questionnaire is **reliable.**

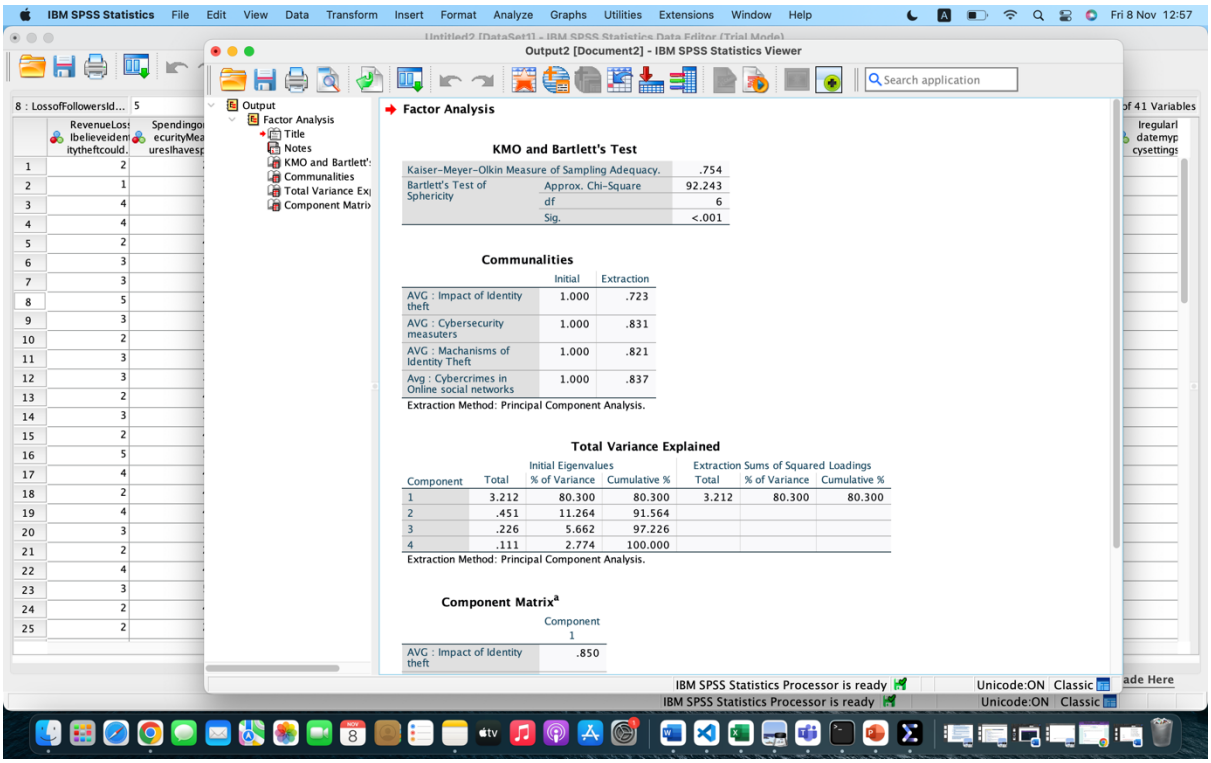
- to check the validity test - KMO and Bartlett's test

Steps in validity analysis

*Analysis >>> Dimension Reduction >>> Facto







KMO

*If KMO value is Greater than 0.7 - Questionnaire is **acceptable**

Bartlett's test - in here has **Sig Value(Significant value)**

*If Sig Value is less than 0.5 - Questionnaire is **acceptable**

In here KMO value is .754. So KMO value is Greater than 0.7 then Questionnaire is acceptable.

In here Bartlett's testing Sig Value is <.001 . So Sig Value is less than 0.5 So Questionnaire is acceptable

b) Main survey (97)

01. Normality test

Steps

*Analysis >>> Descriptive statistics >>> Explore

this diagram must be normal diagram

*** Sample >= 50 ——— Kolmogorov Smirnov test**

*** Sample <= 50 ——— Shapiro Wilk Test**

After doing the **Shapio Wilk** Test we can have a "p value".

* If p value >0.05 then it is Normal

* If p value <0.05 then it is non-normal

02. Correlation analysis

* In here we can have "r value" and "Sig value"

* This r value must be between -1 and +1

* This sig value (2-tailed)must be less than 0.05

——— If your **Normality test is normal** then it is r**unning in the Pearson Correlation**

Steps

*Analysis >>> Correlate >> Bivariate

*In here we can have “r value” and “Sig value”

- This **r value** must be between **-1 and +1**

- This **sig value (2-tailed)**must be l**ess than 0.05**

——— If your **sample is normal** , then we do the **One sample t-test** (it done to reject null hypothesis)

* **Sig(2-tailed) column value** is **close to 0.00** then we **can reject null hypothesis (Ho)**

Steps

* Analysis >>> Compare Means >> One-Sample T-Test

——— If your **Non-Normality test is normal** then it is running in the **Spearman’s RHO**

Steps

* Analysis >>> Correlate >> Bivariate

*In here we can have “***r value***” and “***Sig value***”

- This ***r value*** must be between ***-1 and +1***

- This ***sig value (2-tailed)*** must be ***less than 0.05***

Normal — Regression(linear) Anlaysis

S**teps in** Regression(linear) Anlaysis

Analysis >>> Regression >> Linear

There are 3 tables have

Model Summry

There are 2 values have - R and R^^2

R—— degree of correlation . Usually it is the positive value

R^^2 ——— how well the independent explains the dependent . This has to be grater than 0.3

ANOVA Table

There is a p value . It has to be less then 0.5

Coefficients table

There is the colunm “unstandarzed B” . It must be positive

There is the colunm “sig value” . It must be less than 0.05

me defame ok nm thama H1 vage objectives accepted venava naththam rejected venava.

Non-normal - GLM (Generalised Linear model)

S**teps in** Regression(linear) Analysis

Analysis >>> Genatalized linear models >>> Generalised Linear Models

There is an Sig value .

Sig value >=0.05 accept Ho

Sig value <0.05 accept HA